Computational Pathology and Benchmark Datasets: An Introduction to the Domain and Data

Yannik Queisler

July 15, 2026

Abstract

This report introduces the domain of computational pathology and the four public histopathology datasets it covers. It first explains what diagnostic pathology is, how a stained glass slide becomes a gigapixel whole-slide image, why such images are tiled into patches, and how weakly supervised and frozen foundation-model pipelines turn those patches into predictions. It then describes, in detail, four H&E datasets — CAMELYON16, TCGA-UT, BRACS, and PANDA — presented in the order of the diagnostic question they pose, from detection (*is cancer present?*) through cancer typing and lesion subtyping to ordinal grading (*how aggressive is it?*). Each dataset is given with its clinical background, an example patch, a summary table, and its class distribution.

Contents

1 Computational Pathology

Diagnostic pathology is the branch of medicine that examines tissue under the microscope to detect and characterise disease. When cancer is suspected, a biopsy or surgical resection is fixed, thinly sectioned, mounted on a glass slide, and stained — most commonly with hematoxylin and eosin (H&E), which colours cell nuclei blue-purple and cytoplasm and stroma shades of pink — so that a pathologist can read cellular and tissue morphology and render a diagnosis. This reading is the clinical reference standard for cancer detection, typing, and grading, and it drives treatment decisions. It is also labour-intensive, requires scarce expertise, and is subject to inter- and intra-observer variability, which motivates computational assistance (Madabhushi and Lee, 2016; Song et al., 2023).

Computerised analysis of microscopy images is not new: the earliest work on cell and chromosome image analysis dates back more than fifty years. What changed recently is scale and method. *Digital pathology* digitises the glass slide with a whole-slide scanner, enabling telepathology, archiving, and education; *computational pathology* (CPath) goes further, treating the resulting images as high-dimensional data for quantitative, reproducible analysis and integrating them with clinical and molecular context. A useful distinction here is between the *ground truth*, the abstract label one wishes to predict, and the *gold standard*, the practical benchmark used to approximate it — typically a pathologist's own assessment. Because algorithmic outputs can eventually prove more reproducible than the human benchmark that trained them, CPath must navigate a *gold-standard paradox* in which the reference itself is imperfect (Abels et al., 2019; Madabhushi and Lee, 2016; Hosseini et al., 2024).

The clinical value of CPath is usually framed along two axes. *Automation* standardises tasks pathologists already perform — detecting metastases, subtyping and grading tumours, quantifying immunohistochemical markers — reducing workload and observer variability. *Discovery* identifies signatures beyond human visual limits, such as predicting prognosis, therapy response, or molecular alterations (for example microsatellite instability) directly from H&E morphology; these are the "sub-visual" features that quantitative analysis can mine but a human eye cannot reliably see (Song et al., 2023; Madabhushi and Lee, 2016). Table 1 summarises representative tasks. The datasets in this report all sit on the automation axis: each carries a diagnostic label a pathologist established, and the task is to predict it from H&E tissue.

Role	Representative tasks
Automation	Metastasis detection; cancer-type classification; tumour subtyping; Gleason/ISUP grading; mitosis counting; immunohistochemistry (PD-L1, Ki-67) quantification
Discovery	Prognosis and recurrence-risk prediction; molecular-marker prediction (e.g. microsatellite instability) from H&E; therapy-response biomarkers

Table 1: Two roles of computational pathology, following Song et al. (2023) and Madabhushi and Lee (2016). The datasets in this report are automation tasks with pathologist-established labels.

1.1 The whole-slide imaging pipeline

A digitised slide is a *whole-slide image* (WSI): a single H&E image of an entire tissue section, routinely on the order of 100,000×100,000 pixels and 0.5–4 GB in size. To make this tractable, a WSI is stored as a *pyramid* of progressively downsampled copies, so the same slide can be read at several magnifications (Table 2): high magnification resolves nuclear detail and mitoses, while lower magnifications carry the tissue-architectural context that diagnosis also depends on. Scanning the whole section, rather than a manually chosen field of view, removes selection bias and captures intratumoral heterogeneity, but it produces far more data than fits in GPU memory (Song et al., 2023; Abels et al., 2019).

Magnification	Resolution	What it resolves
40×	≈0.25 μm/pixel	Nuclear detail, chromatin, mitotic figures
20×	≈0.5 μm/pixel	Cellular morphology, individual glands
10×	≈1.0 μm/pixel	Local tissue architecture
5×	≈2.0 μm/pixel	Macro-architectural context

Table 2: Typical magnification levels of a pyramidal WSI and the morphology each makes legible (Song et al., 2023). High magnifications are needed for nuclear grading, lower ones for architectural context.

The standard response to the size of a WSI is a divide-and-conquer strategy: the image is partitioned into small *patches* (also called *tiles*), 256 × 256 pixels to match the input size of the Virchow2 encoder used throughout (Section 1.2), each processed independently and later aggregated into a slide- or patient-level prediction (Song et al., 2023). How each dataset arrives at those tiles differs — from author-supplied patches to on-the-fly tiling of raw slides — and the magnification at which they are cut varies too; each dataset below states its own tiling. Figure 1 sketches the resulting pipeline, from stained slide to prediction, that every dataset in this report passes through.

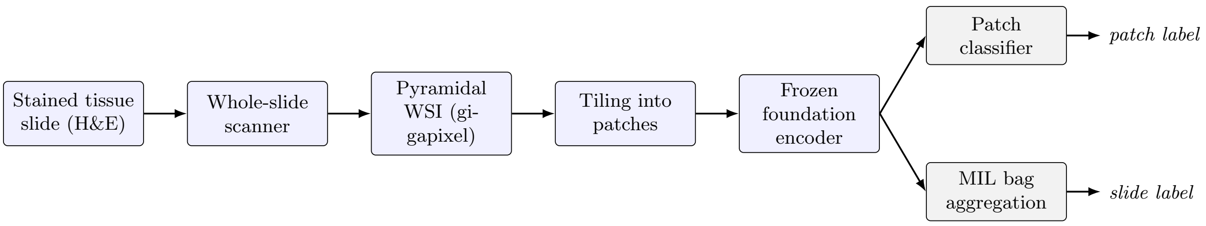


Figure 1: The computational-pathology pipeline the datasets pass through. A stained slide is scanned into a pyramidal WSI and tiled into patches; a frozen foundation-model encoder embeds each patch once, after which either a patch classifier predicts a per-tile label or a multiple-instance head aggregates the patch features of a slide into a slide-level prediction.

1.2 Learning from patches: weak supervision and foundation models

How labels attach to these units determines the learning regime. When each patch carries a reliable label — because it was cropped from a pathologist-selected region, or intersected with an expert pixel mask — ordinary patch classification applies. When only a slide-level diagnosis is available, the problem is *weakly supervised*: the slide is a *bag* of patch *instances* that share one label, and *multiple-instance learning* (MIL) aggregates the instances — typically through a learned attention mechanism that weights diagnostically salient patches — into a slide prediction (Song et al., 2023; Hosseini et al., 2024). Because tiling discards spatial context, context-aware models such as graph neural networks (over patches or nuclei) and transformers (with global self-attention) are sometimes used to recover it, but the attention-MIL bag remains the workhorse for slide-level tasks. Both regimes appear in the datasets below, and two of them (CAMELYON16 and PANDA) support both on the same slides.

The features fed to these heads have shifted from hand-crafted descriptors — interpretable measurements of nuclear shape, texture, or spatial architecture — to learned representations that capture patterns beyond human perception. Modern CPath rarely trains an image encoder from scratch: a pathology *foundation model*, pre-trained self-supervised on large slide archives, provides a frozen feature extractor that embeds each patch once into a fixed-length vector, leaving only a lightweight classifier or MIL head to train. Virchow2 is one such H&E foundation model (Vorontsov et al., 2024; Zimmermann et al., 2024). Freezing the representation lets a study isolate the effect of a downstream modelling choice from the confounds of end-to-end backbone training, and makes the patch and MIL regimes directly comparable, because both consume the same embeddings.

1.3 Data and annotation

Deep learning is data-hungry, and expert annotation is the binding constraint in CPath: pathologist labelling is slow, expensive, and hard to scale, so datasets range from densely mask-annotated cohorts to those carrying only a single slide-level diagnosis (Abels et al., 2019; Hosseini et al., 2024). The four datasets in this report deliberately span that range of annotation form, from pixel-level tumour masks through region-of-interest crops to a single slide-level diagnosis. The next section introduces them in the order of the diagnostic question they pose.

2 Datasets

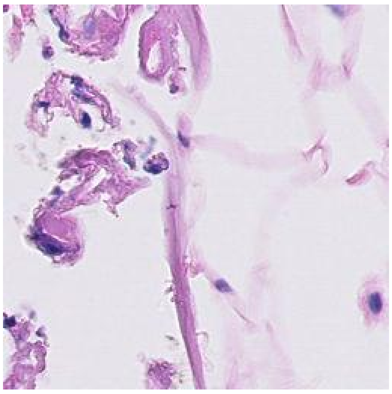
Four public H&E datasets are used, described here in the order of the diagnostic question they pose — an arc that also escalates the structure of the label and varies the organ and the form of annotation:

- **CAMELYON16** — *is cancer present?* Binary metastasis *detection* in sentinel lymph nodes, with pixel-mask annotations.
- **TCGA-UT** — *what cancer is it?* Pan-cancer *typing* across 32 organs, each patch labelled with its slide's cancer type.
- **BRACS** — *how abnormal is this tissue?* Breast-lesion *subtyping* along the benign–atypical–malignant continuum, with region-of-interest annotations.
- **PANDA** — *how aggressive is it?* Ordinal ISUP *grading* of prostate biopsies, with region masks and built-in two-provider heterogeneity.

2.1 CAMELYON16 — lymph-node metastasis detection

CAMELYON16 is an organ-specific H&E cohort of sentinel lymph-node sections for the detection of breast-cancer metastases (Ehteshami Bejnordi et al., 2017; Litjens et al., 2018). The diagnostic question is the most basic in oncology — *is cancer present?* — and the label is binary: each slide is either *normal* or contains *tumor* (metastasis). Expert annotations are supplied as pixel-level masks marking tumour tissue, rather than as cropped regions, so individual tiles can be labelled by intersecting them with the mask. The slides are tiled once into non-overlapping 256×256 patches at 20x by the group's preprocessing pipeline, which keeps a patch only if it clears a tissue-foreground stack — Otsu thresholding (at least 10% foreground), a standard-deviation filter (at least 8 gray levels), Canny edge detection, and an isolation filter that discards patches with fewer than two tissue neighbours. Each surviving patch is then labelled from the WSI mask (value 1 normal, 2 tumour).

The benchmark uses 398 of the 399 published slides (160 tumor, 238 normal) after excluding one slide without usable tile evidence; each slide is its own patient, so slide-disjoint splits coincide with patient-disjoint splits. Because metastatic tissue occupies only a small fraction of a sectioned node, mask-derived tumor tiles are scarce: the patch distribution is 2,340,040 normal against 59,672 tumor tiles (Table 3, Figure 2), even though tumour and normal slides are themselves close to even (160 versus 238)—tumour tiles are scarce even on slides that do contain a metastasis. The 20 tumor slides whose annotations are not exhaustive are excluded from the tile-labelling regime, where unannotated tumour could be mislabelled as normal, but retained where only the reliable slide-level label is used.



Property	Value
WSIs (tumor / normal)	160 / 238
Patch-labelled WSIs	378
Patch labels	tumor, normal
Eligible patches (tumor / normal)	59,672 / 2,340,040

Table 3: CAMELYON16 dataset properties, alongside an example 256 × 256 tumor patch. Tumour patches are scarce even within slides that contain a metastasis.

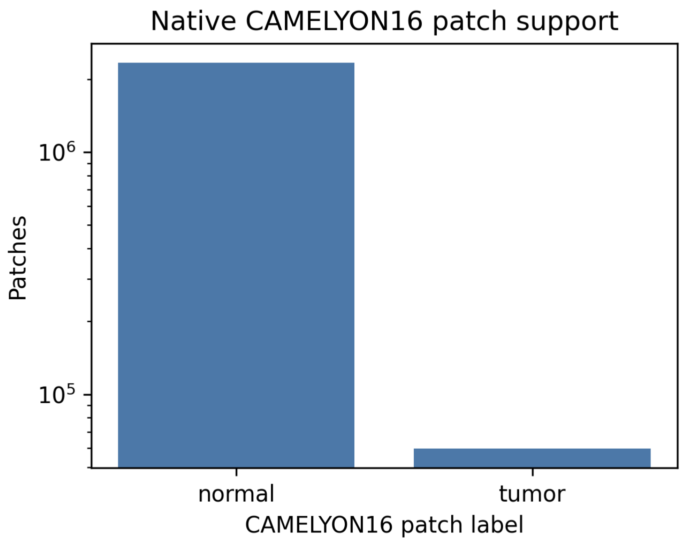
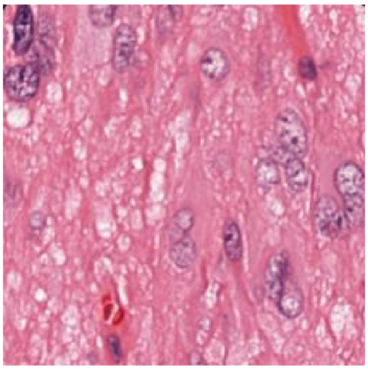


Figure 2: CAMELYON16 patch support by tumor/normal label after mask-based labelling (log scale). Tumor tiles are far scarcer than normal tiles.

2.2 TCGA-UT — pan-cancer typing

TCGA-UT widens the lens from one organ to the whole body. It is derived from diagnostic WSIs in The Cancer Genome Atlas (TCGA), a pan-cancer resource linking molecular and clinical data with large image collections (Tomczak et al., 2015). The TCGA-UT release turns this slide archive into a computational-pathology dataset by cropping H&E tumour patches from pathologist-selected uniform tumour regions: 1,608,060 RGB patches of 256 × 256 pixels from 8,736 diagnostic slides, across 32 cancer types (Komura et al., 2022). These patches are not tiled here but supplied by the dataset authors: for each slide at least three uniform tumour regions were delineated as polygons by trained pathologists, and 256×256 patches were extracted from them across six magnifications (0.5–1.0 μm/px) and resized to 256×256 (Komura et al., 2022). No tissue-foreground filter is applied, the regions already being curated tumour. Every patch carries the cancer-type label of the slide it was cropped from, so the patches are themselves labelled samples — the unit of patch-level classification — while their slide identities also allow grouping into slide-level bags. The labels are cancer types, not dense pixel masks or cell-level annotations, so the diagnostic question is *what cancer is it?*, a nominal classification across 32 organs.

The largest class is glioblastoma multiforme (GBM; 792 slides) and the smallest is diffuse large B-cell lymphoma (DLBC; 28 slides), with the remaining types spread across a gradually declining tail (Table 4, Figure 3).



Property	Value
Cancer-type labels	32
Diagnostic slides	8,736
Author-supplied patches	1,608,060 of 256×256
Largest class	Glioblastoma (GBM), 792 slides
Smallest class	B-cell lymphoma (DLBC), 28 slides

Table 4: TCGA-UT dataset properties, alongside an example 256×256 H&E tumor patch. Because every patch inherits its slide's cancer-type label, the patch- and slide-level class distributions coincide.

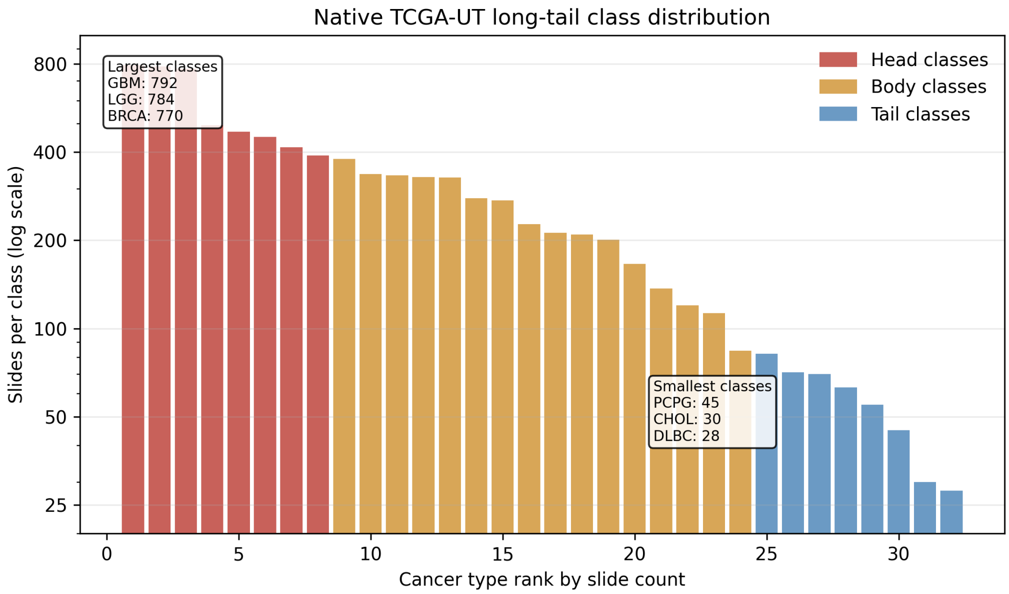
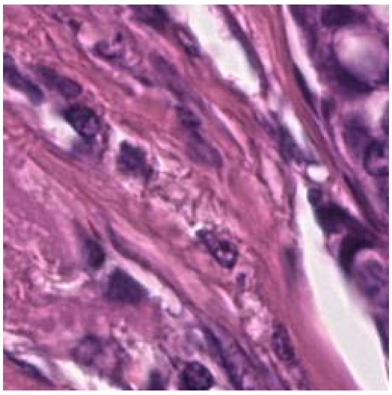


Figure 3: TCGA-UT slide support by cancer-type rank, declining gradually across the 32 classes from GBM at the head to DLBC at the tail.

2.3 BRACS — breast-lesion subtyping

BRACS narrows back to a single organ but sharpens the question from *which cancer* to *how abnormal is this tissue*. It is an H&E breast-pathology dataset of 547 WSIs and 4,539 regions of interest (ROIs) from 189 patients, annotated by consensus of three board-certified pathologists (Brancati et al., 2022). The labels form three lesion types — benign, atypical, and malignant — refined into seven subtypes: normal (N), pathological benign (PB), usual ductal hyperplasia (UDH), flat epithelial atypia (FEA), atypical ductal hyperplasia (ADH), ductal carcinoma in situ (DCIS), and invasive carcinoma (IC). Annotations are ROI crops rather than whole-slide labels or pixel masks, and a single slide may contain ROIs of several subtypes.

The benchmark uses the seven-subtype target, which keeps clinically important but comparatively scarce precancers visible: the atypical lesions FEA and ADH would be hidden by collapsing to three lesion types. The pathologist-drawn ROIs are supplied as `.png` crops at the slides' native 0.25 μm/px (40x) resolution; each is gridded into non-overlapping 256×256 tiles with no tissue-foreground filter, the ROIs already being tissue. This yields 278,028 tiles from the 4,539 ROIs of 387 WSIs (151 patients); 16 ROIs are too small for even one tile. Counting the slides that contain each subtype (a slide with several lesions counts under each), support ranges from PB (196 slides) at the head to DCIS (87 slides) at the tail (Table 5, Figure 4). BRACS's difficulty comes from fine-grained morphology rather than a steep tail.



Property	Value
Patients	151
ROI-bearing WSIs	387
ROI-derived patches	278,028
Subtype labels	N, PB, UDH, FEA, ADH, DCIS, IC

Table 5: BRACS dataset properties, alongside an example 256×256 ROI patch. Slide-level support counts each slide under every subtype it contains; the patch-level distribution instead weights subtypes by their patch counts.

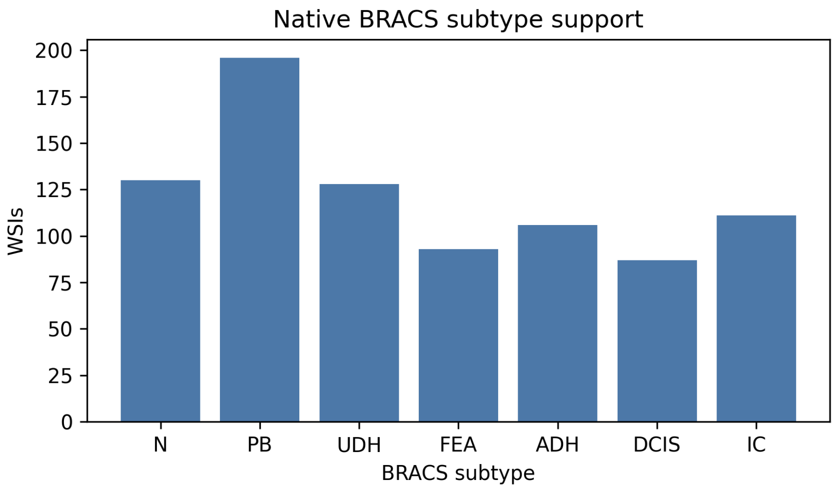


Figure 4: BRACS WSI support by seven-subtype label (slides containing each subtype), from PB at the head to DCIS at the tail.

2.4 PANDA — prostate cancer grading

PANDA (Prostate cANcer graDe Assessment) poses the most nuanced question — *how aggressive is the cancer?* — as an ordinal grading problem. It is an H&E dataset of prostate core-needle biopsies assembled for automated Gleason grading and the largest public cohort of its kind, with 10,616 whole-slide biopsies (Bulten et al., 2022). Gleason grading describes prostate-cancer severity from glandular growth patterns; pathologists summarise the dominant and secondary malignant patterns into ordinal ISUP Grade Groups, where higher groups indicate more aggressive morphology. Here grade 0 marks benign tissue and grades 1–5 are the ordered ISUP cancer groups.

PANDA differs from the other three along every axis a generality check exercises: a third organ (prostate), an ordinal rather than nominal or binary label, and a built-in source of heterogeneity, because the slides come from two institutions — Radboud University Medical Center and Karolinska Institute — that differ in staining, scanning, and annotation convention, and grade differently (Karolinska biopsies concentrate at low grades, Radboud spread more evenly). Across the full cohort of 10,616 biopsies (Karolinska 5,456, Radboud 5,160), the ISUP distribution runs from ISUP 0 (2,892 biopsies) at the head to ISUP 5 (1,224) at the tail — a mild $\approx$2.4:1 slide-level imbalance.

Unlike the others, PANDA ships raw pyramidal biopsies rather than ready-made patches, so tiles are cut here: each biopsy is read at level 0, its native full resolution, divided into 256×256 tiles, and a luminance foreground filter keeps every tile that is at least 35% tissue (mean intensity below 210), discarding glass and background. No per-slide cap is applied, so the counts reflect each biopsy's full available tissue. PANDA's masks additionally support a provider-consistent binary cancer/benign tile label — the only labelling the two institutions share — formed by collapsing the provider-specific mask codes and calling a tile *cancer* when at least half of its mask cell carries a cancer value. Over the 10,514 masked biopsies that yield labelled tissue this gives 3,702,544 benign against 803,785 cancer tiles, so the same biopsies furnish both an ordinal slide-level grading target and a binary tile-level detection target (Table 6, Figure 5).

PANDA is the only one of the four datasets whose two regimes answer different questions, and this follows from its annotations rather than from a design choice. It is the one cohort carrying two independent label sources at two granularities: a slide-level ordinal ISUP grade, which summarises the whole biopsy and cannot be assigned to a single tile, and pixel masks that yield a per-tile label. CAMELYON16, TCGA-UT, and BRACS instead each provide a single label concept — tumour/normal, cancer type, or lesion subtype — that applies unchanged at both tile and slide level, so their two regimes share one task and differ only in prediction unit.

Property	Value
WSIs (biopsies)	10,616
Patch-labelled WSIs	10,514
Slide labels / patch labels	ISUP 0–5 / benign, cancer
Eligible patches (benign / cancer)	3,702,544 / 803,785

Table 6: PANDA dataset properties from full-cohort tiling and mask-based patch labelling. Unlike the other datasets, patch and slide labels define different tasks: binary cancer detection and six-class ISUP grading, respectively.

3 Summary

This report introduced the computational-pathology pipeline — from a stained glass slide to a gigapixel whole-slide image, patching, weakly supervised multiple-instance learning, and frozen foundation-model features — and the four H&E datasets it covers, presented as a progression of diagnostic questions from CAMELYON16 detection through TCGA-UT typing and BRACS subtyping to PANDA grading, each with its organ, label type, annotation form, example patch, and class distribution.

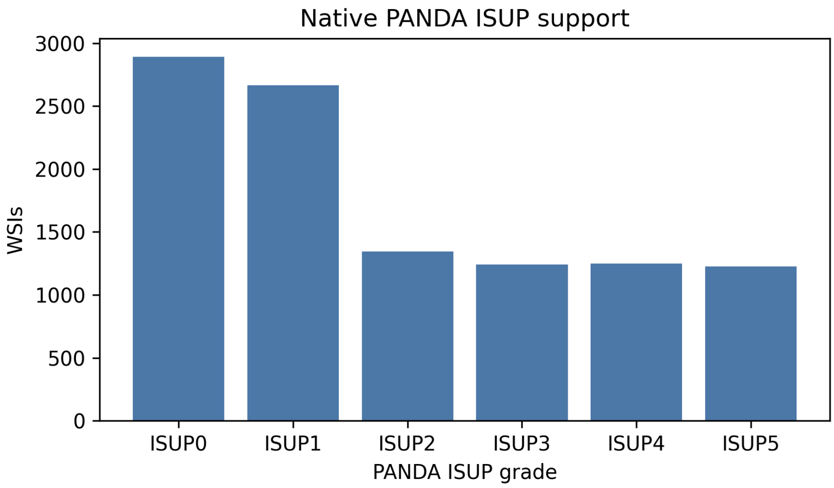


Figure 5: PANDA WSI support by six-class ISUP grade across the full 10,616-biopsy cohort, from ISUP 0 at the head to ISUP 5 at the tail.

References

Esther Abels, Liron Pantanowitz, Famke Aeffner, Mark D. Zarella, and Jeroen van der Laak. Computational pathology definitions, best practices, and recommendations for regulatory guidance: a white paper from the digital pathology association. *The Journal of Pathology*, 249(3):286–294, 2019. doi: 10.1002/path.5331.

Nadia Brancati, Anna Maria Anniciello, Pushpak Pati, Daniel Riccio, Giosue Scognamiglio, Guillaume Jaume, Giuseppe De Pietro, Maurizio Di Bonito, Antonio Foncubierta, Gerardo Botti, Maria Gabrani, Florinda Feroce, and Maria Frucci. BRACS: A dataset for BReAst carcinoma subtyping in H&E histology images. *Database*, 2022:baac093, 2022. doi: 10.1093/database/baac093.

Wouter Bulten, Kimmo Kartasalo, Po-Hsuan Cameron Chen, Peter Ström, Hans Pinckaers, Kunal Nagpal, Yuannan Cai, David F. Steiner, Hester van Boven, Robert Vink, Christina Hulsbergen-van de Kaa, Jeroen van der Laak, Mahul B. Amin, Andrew J. Evans, Theodorus van der Kwast, Robert Allan, Peter A. Humphrey, Henrik Grönberg, Hemamali Samaratunga, Brett Delahunt, Toyonori Tsuzuki, Lars Egevad, Geert Litjens, Martin Eklund, and the PANDA challenge consortium. Artificial intelligence for diagnosis and Gleason grading of prostate cancer: the PANDA challenge. *Nature Medicine*, 28(1):154–163, 2022. doi: 10.1038/s41591-021-01620-2.

Babak Ehteshami Bejnordi, Mitko Veta, Paul Johannes van Diest, Bram van Ginneken, Nico Karssemeijer, Geert Litjens, Jeroen A. W. M. van der Laak, and the CAMELYON16 Consortium. Diagnostic assessment of deep learning algorithms for detection of lymph node metastases in women with breast cancer. *JAMA*, 318(22):2199–2210, 2017. doi: 10.1001/jama.2017.14585.

Mahdi S. Hosseini, Danial Hasan, Xingwen Li, Stephen Yang, et al. Computational pathology: A survey review and the way forward. *Journal of Pathology Informatics*, 15:100357, 2024. doi: 10.1016/j.jpi.2023.100357.

Daisuke Komura, Akihiro Kawabe, Kenta Fukuta, Kazuya Sano, Tsukasa Umezaki, Haruka Koda, Riku Suzuki, Keisuke Tominaga, Michiaki Ochi, Hirofumi Konishi, et al. Universal encoding of pan-cancer histology by deep texture representations. *Cell Reports*, 38(9):110424, 2022. doi: 10.1016/j.celrep.2022.110424.

Geert Litjens, Peter Bandi, Babak Ehteshami Bejnordi, Oscar Geessink, Maschenka Balkenhol, Peter Bult, Altuna Halilovic, Meyke Hermsen, Rob van de Loo, Rob Vogels, Quirine F. Manson, Nikolas Stathonikos, Alexi Baidoshvili, Paul van Diest, Carla Wauters, Marcory van Dijk, and Jeroen van der Laak. 1399 H&E-stained sentinel lymph node sections of breast cancer patients: the CAMELYON dataset. *GigaScience*, 7(6):giy065, 2018. doi: 10.1093/gigascience/giy065.

Anant Madabhushi and George Lee. Image analysis and machine learning in digital pathology: Challenges and opportunities. *Medical Image Analysis*, 33:170–175, 2016. doi: 10.1016/j.media.2016.06.037.

Andrew H. Song, Guillaume Jaume, Drew F. K. Williamson, Ming Y. Lu, Anurag Vaidya, Tiffany R. Miller, and Faisal Mahmood. Artificial intelligence for digital and computational pathology. *Nature Reviews Bioengineering*, 1(12):930–949, 2023. doi: 10.1038/s44222-023-00096-8.

Katarzyna Tomczak, Patrycja Czerwinska, and Maciej Wiznerowicz. The cancer genome atlas (TCGA): an immeasurable source of knowledge. *Contemporary Oncology*, 19(1A):A68–A77, 2015. doi: 10.5114/wo.2014.47136.

Eugene Vorontsov, Alican Bozkurt, Adam Casson, George Shaikovski, Michal Zelechowski, Kristen Severson, Eric Zimmermann, et al. A foundation model for clinical-grade computational pathology and rare cancers detection. *Nature Medicine*, 30(10):2924–2935, 2024. doi: 10.1038/s41591-024-03141-0.

Eric Zimmermann, Eugene Vorontsov, Julian Viret, Adam Casson, Michal Zelechowski, George Shaikovski, Neil Tenenholtz, James Hall, Thomas Fuchs, Nicolo Fusi, Siqi Liu, and Kristen Severson. Virchow2: Scaling self-supervised mixed magnification models in pathology. *arXiv preprint arXiv:2408.00738*, 2024. doi: 10.48550/arXiv.2408.00738.